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Basic characteristics of a multilayer thin electrostatic actuator supported by lubricating oil for a fine-motion stage

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ABSTRACT

The present paper describes the basic characteristics of a multilayer thin electrostatic actuator supported by lubricating oil rather than precise ball bearings for a fine-motion stage. The purpose of the present research is to realize an easy-to-maintain multilayer thin electrostatic actuator having a simple structure for high-precision motion. Lubricating oil is used to support the mover, which is constructed of thin electrodes, and to reduce the friction between the mover and stator layers without the need for precise ball bearings. Each layer of the actuator has only one simple electrode that comprises several beams. In order to examine the characteristics of the multilayer thin electrostatic actuator, a two-layer electrostatic actuator was designed and constructed. First, the effect of the viscosity of the lubricating oil on the displacement and force characteristics were examined in order to clarify the suitable lubricating oil characteristics. In the experiments, low-viscosity oil provided a faster response and a larger thrust force. Next, a suitable driving signal for the electrostatic actuator supported by lubricating oil was investigated experimentally. Based on the results, control systems for the electrostatic actuator using suitable driving signals were designed in order to examine the effectiveness of the actuator. The positioning results indicate that the positioning accuracy is better than 15 nm.

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1. Introduction

Fine-motion stages are widely used in semiconductor manufacturing systems, machine tools, and scanning probe microscope systems, which often require high positioning accuracy in a short positioning time [1–3]. It is also preferred that fine-motion stages generate low heat, have a simple structure, and be easy to maintain. At present, piezoelectric actuators [4–11] and electromagnetic actuators [14–20] are used in a number of fine-motion stages. Table 1 shows the general characteristics of the actuators. Piezoelectric actuators have the ability to produce large thrust forces [5,8] and generally generate very little heat [4]. However, the disadvantages are that they often require hinges for multiple degree of freedom (DOF) motion [4–11] and for extension of their short working range of less than 10 μm [4]. The addition of hinges makes the structure complex and increases the size of the stage. Furthermore, hinges become vibration elements, which will increase the design complexity of the system [4,5]. In comparison, electromagnetic actuators can work with multi-DOF fine-motion stages without hinges, produce a large thrust force and are capable of long working range motion [14–17]. Electromagnetic actuators, however, generate large amount of heat that may deform products on the stages as well as the fine-motion stages themselves which is a disadvantage [2,14].

In contrast to the previous discussed actuators, electrostatic actuators produce little heat and can be produced without rare metals or rare earth materials. In addition, unlike piezoelectric actuators, electrostatic actuators can transmit power without mechanical contact and do not require hinges. However, in micro-electro-mechanical systems, comb-drive type electrostatic actuators are often used in conjunction with elastic hinges [12,13]. The use of elastic hinges makes the design and fabrication processes of the system complex. In general, their working range is limited to less than 10 μm [12] and often too short for fine motion stages. Although some comb-drive type electrostatic actuators have a working range of 100 μm, they require the implementation of a more complicated hinge mechanism [13].

Variable-capacitance motors [21–26,29–31] and induction motors [27,28] type electrostatic actuator do not have elastic hinges and enable longer working range motion exceeding 200 μm. In general, electrostatic actuators produce small thrust force. However, a multilayer structure enables the actuator to produce large thrust force. In these actuators [21–23,27,28], Higuchi et al. implemented precise beads as rolling elements of a ball guide in order to reduce the frictional effect between electrodes without groves for guiding the beads. The prototype actuators show high thrust density. However, copious precise beads must be spread on the electrode surface in order to maintain the gap and low-friction character-

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Table 1
General characteristics of actuators.

	Characteristics				
	Working range	Transmit power without mechanical contact	Heat dissipation	Thrust force	Structure
Piezoelectric actuator	Short	No	Relatively low	Large	Complex
Electromagnetic actuator	Long	Yes	High	Large	Complex
Electrostatic actuator	Long	Yes	Low	Small	Simple

istics. Ghodssi et al. [24–26] employed microball-bearings with a 3D guide structure fabricated on the mover and stator layers in order to support the microball-bearings. Although the guide structure enables several microballs to support the mover, the structures require complex fabrication process and needs to be provided on all the layers. As a result, the use of both precise beads and microball bearings will increase cost. In industrial applications, a simple structure and ease of maintenance are extremely important.

In the electrostatic actuator, a mover supported by lubricating oil is not affected by the above problems, which are associated with the use of precise beads or microball-bearings. In conventional stages, support by lubricating oil often causes notable friction force when supporting heavy objects. However, a multilayer electrostatic actuator is lightweight and is expected to cause small friction force by the mover mass. The thin mover layers are also expected to float due to the capillary force of the lubricating oil. The purpose of the present study is to realize an easy-to-maintain multilayer thin electrostatic actuator having a simple structure supported by lubricating oil for a precise and fine-motion stage. For this purpose, a two-layer electrostatic actuator supported by lubricating oil was constructed and evaluated. In the present paper, the basic characteristics of the actuator are examined, and a suitable driving and control method is proposed.

In Section 2, the design of the actuator and the experimental setup are described. In Section 3, the effects of oil viscosity on open-loop characteristics are examined in order to clarify suitable oil characteristics. In Section 4, the effect of driving signals on motion characteristics are examined experimentally in order to determine a suitable driving signal for an actuator supported by lubricating oil rather than precise ball bearings. In Section 5, a driving signal was applied to the control structure of the actuator in order to ensure the full working range of the actuator and demonstrate the control performance of the actuator. Conclusions are presented in Section 6.

2. Experimental electrostatic actuator

The electrostatic actuator considered in the present study is a variable-capacitance motor. Fig. 1 shows a schematic diagram of the experimental two-layer electrostatic actuator to examine the characteristics of a multi-layer thin electrostatic actuator supported by lubricating oil. Having numerous parallel beams, the thin electrodes of the stator and the mover are alternately stacked. The mover mass is 5.22 g and the actuator can support the mass by lubricating oil without applied voltages. This structure is free from dielectric breakdown in each layer and the beam pitch is easy to adjust. For bi-directional motion, two voltage signals, namely, V_1 and V_2, are alternately applied to Stator A and Stator B, while the mover is set to zero (grounded). In this actuator, the working range is 500 μm, as determined by the beam pitch. For maintaining the gap between the electrodes, lubricating oils are used without precise ball bearings because the use of the lubricating oils make the fabrication and maintenance process easier. The electrodes are covered with ethylene thetrafluoroethylene (ETFE) flat films as isolation films. The ETFE films are adopted in order to keep the frictional effect low, even if the electrodes contact each other.

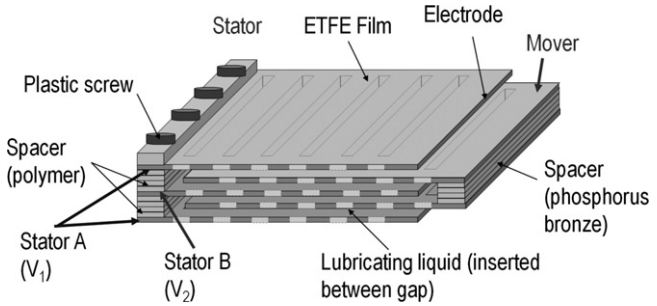


Fig. 1. Schematic diagram of an experimental two-layer electrostatic actuator.

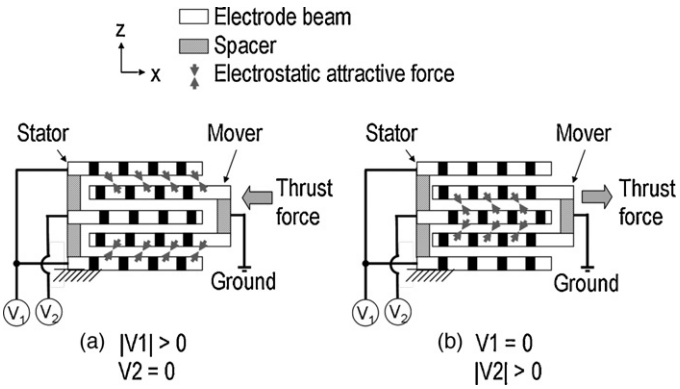


Fig. 2. Driving procedure for the bi-directional motion of the electrostatic actuator.

Fig. 2 shows the driving method for the bi-directional motion of the electrostatic actuator. When the absolute value of voltage V_1 is not equal to zero and voltage V_2 is equal to zero, the mover produces a thrust force toward the left due to the electrostatic attractive force between the electrodes (see Fig. 2(a)). As shown in Fig. 2(b), when the absolute value of voltage V_2 is not equal to zero and voltage V_1 is equal to zero, the mover produces a thrust force toward the right. The applied voltages to the electrostatic actuator produce not only the thrust force in the x-direction but also the attractive force between the electrodes in the z-direction. The influence of the attractive force on the generated thrust force in the x-direction is discussed in Section 3.2. The dynamic model of the actuator can be expressed as shown in Fig. 3. Its parameters are shown in Table 2.

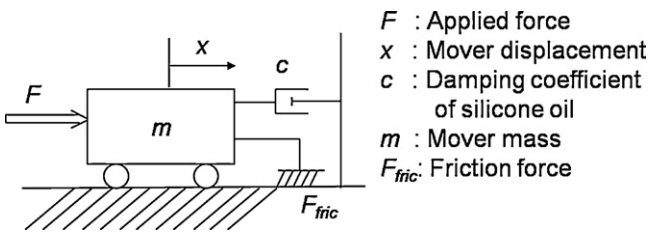


Fig. 3. Dynamic model.

Table 2
Specifications of the electrostatic actuator.

Parameters	Value
Electrode width and length (mm)	50 × 50
Beam pitch (mm)	1.5
Spacer (mm)	50 × 6.5
Electrode thickness (mm)	0.1
Spacer thickness, four pieces (mm)	0.4
ETFE film thickness (mm)	0.014
Electrode + ETFE film thickness (mm)	0.128
Thickness of assembled mover (mm)	0.656
Thickness of assembled stator (mm)	1.184
Gap between electrodes (mm)	0.244
Estimated silicon oil thickness (mm)	0.244
Mover mass (g)	5.22

This dynamic model is used for thrust force estimation in Section 3.

The basic displacement and force characteristics of the actuator are first examined. Fig. 4 shows the displacement measurement setup with a capacitance displacement sensor. The two-layer electrostatic actuator (see Fig. 1) is also shown in Fig. 4. The measurement range of the sensor is 50 μm, and the evaluation range of the actuator characteristics is smaller than 50 μm. Thus, in the present study, the full working range motion characteristics are examined within a range of 40 μm. For measurement of the force characteristics, the capacitance displacement sensor is replaced with a load cell.

3. Open-loop characteristics

In the experimental actuator, silicone oils are used as lubricating oils. In this section, several silicone oils with various viscosities are used to identify the most suitable oil. The viscosity is considered to influence the lubrication characteristics of the actuator and change the displacement and force characteristics. A higher-viscosity silicone oil is expected to be easier to maintain inside the gap with respect to the electrostatic attractive force for low friction. On the other hand, it is expected that a lower-viscosity oil provides lower damping characteristics in the direction of travel for high response.

3.1. Displacement characteristics

Fig. 5 shows the open-loop step responses of the actuator. Fig. 5(a) demonstrates an example of the time response with three silicone oils to a 2-kV step input. The dielectric constants of the 10-, 50-, and 100-mPa s silicone oils are 2.65, 2.72, and 2.74, respectively. For all measured open-loop responses, similar experiments

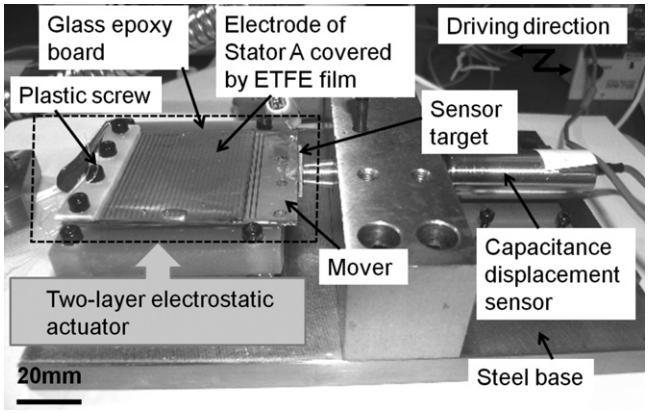


Fig. 4. Overview of the displacement measurement setup.

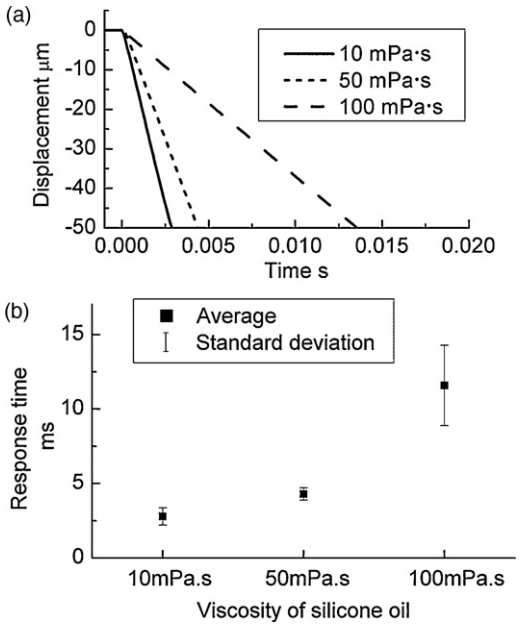


Fig. 5. Open-loop step responses and mean response time using a voltage of 2 kV. (a) Open-loop step responses under different viscosity silicone oil. (b) Average and the standard deviation of the time response under different viscosity silicone oil.

were conducted 20 times. In the present paper, the time required for the 50-μm displacement is referred to as the response time and examined as an index of actuator response. The average and the standard deviation of the response time are shown in Fig. 5(b). In Fig. 5(b), the 10-mPa s silicone oil is seen to exhibit the shortest response time compared to the other viscosity silicone oil.

3.2. Force characteristics

In contrast to gaps maintained by precise balls as parts of ball guides, a gap maintained by lubricating oil is sensitive to an attractive force produced by the applied voltages between the electrodes. When voltage is applied to the actuator, the attractive force between the electrodes will decreases the gap thus increasing the frictional effect. These changes influence the horizontal force characteristics of the actuator. The attractive force may be influenced by the oil viscosity and the signal pattern applied to the electrodes. In this section, three approaches were taken in order to examine their effects on the horizontal force characteristics, i.e., thrust force estimation from the open-loop step displacement response (referred to as the estimated force), thrust force measurement using a step input signal (referred to as the step signal force), and thrust force measurement using a triangular input signal having a period of 1 s (referred to as the quasi-static signal force). The estimated force was determined from curve fitting of the simulated response by Eq. (1) and the dynamic model shown in Fig. 3 in order to measure the response. The minimum voltage was measured by applying a stepwise input voltage to the experimental actuator. The step signal force and the quasi-static signal force were measured with a load cell directly. In the three approaches, the estimated force is expected to be the least sensitive to the attractive force because the period of applied voltages is the shortest. The quasi-static signal force is expected to be strongly influenced by the attractive force because the long period of applied voltage.

$$F_{\text{outputted}} = k(V_{\text{in}}^2 - V_{\text{min}}^2) \quad (1)$$

where $F_{\text{outputted}}$ = thrust force output by the actuator, k = thrust force gain, V_{in} = applied voltage and V_{min} = minimum voltage to move the mover.

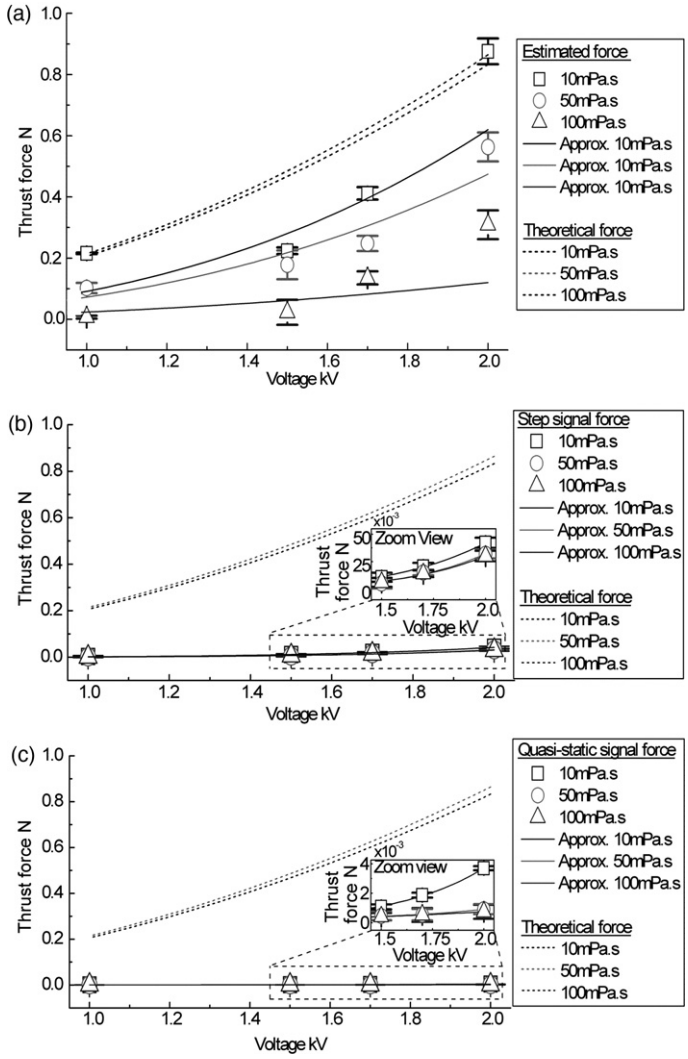


Fig. 6. Relationship between the input voltage and the thrust force. (a) Comparison between estimated force and theoretical force. (b) Comparison between step signal force and theoretical force. (c) Comparison between quasi-static signal force and theoretical force.

Fig. 6 shows the relationship between the input voltage and the thrust force. The three thrust force derivation approaches with three silicone oils are shown in three separate figures. In order to obtain each result, all of the experiments were repeated 20 times. Each symbol indicates the average and the standard deviation of the thrust force under each condition. Each curve represents the approximate curve calculated using the least squares method and Eq. (1).

In Fig. 6, the 10-mPa s silicone oil produces a larger thrust force than the other silicone oils for all of the assessed approaches. This result indicates that the low-viscosity oil is suitable for a large thrust force. Although the high-viscosity oil was expected to reduce the friction effect, no advantage was observed. Therefore, low-viscosity oil is revealed to be suitable for the electrostatic actuator. In addition, the estimated force as shown in Fig. 6(a) is closer to the theoretical thrust force compared to the other thrust forces shown in Fig. 6(b) and (c). In comparison with the measured thrust forces (see Fig. 6(b)and (c)), the displacement of the actuator immediately after the voltage input was used for the estimated force (see Fig. 6(a)). The period of applying voltages for the estimated force is shorter than those for the two measured thrust forces. The comparison results indicate that the longer period of applying voltages to the actuator increases the reduction effect of the gap and the

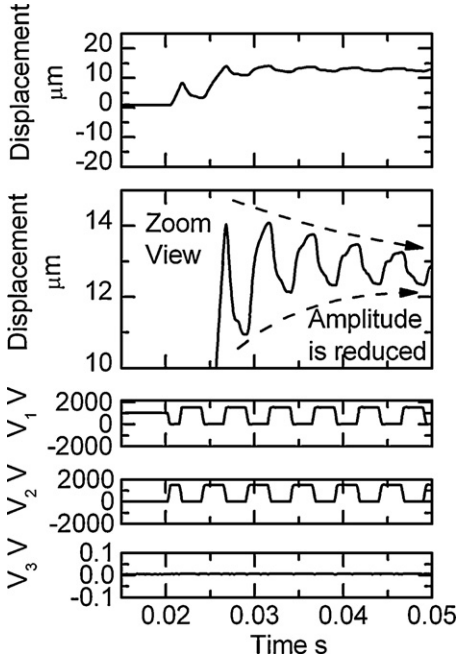


Fig. 7. Displacement response of the actuator to a periodic rectangular input signal using silicone oil with a viscosity of 10 mPa s.

frictional force between the electrodes by the attractive force. As a result, the estimated force is least influenced by the electrostatic attractive force due to the shorter period of applied voltages. The estimated force for 10-mPa s silicone oil is nearly equal to the theoretically produced thrust force by 71.7% as shown in Fig. 6(a). Therefore, for full working range motion, the driving signals, which do not influence the working range, need to be clarified.

4. Driving signals the actuator

4.1. Basic dynamic displacement characteristics

As mentioned in Section 3.2, the long period of applying voltage signal to the actuator greatly reduces the gap between the electrodes and increases the frictional force. As a result, the available thrust force decreases, although the applied voltage produces the thrust force itself. Fig. 7 shows the displacement response of the actuator to a periodic rectangular input signal. In this figure, voltages V_1 and V_2 are applied to Stators A and B, respectively, as shown in Fig. 1, and the mover is grounded. Voltages V_1 and V_2 are out of phase by π. The periodic rectangular input signal has a frequency that is lower than the control bandwidth. In the present paper, the periodic rectangular input signal is referred to as the normal signal. In Fig. 7, the initial displacement is large, but the amplitude of the displacement decreases step by step to approximately 2 μm_{p-p}. This indicates that the period of the input signal is long enough to produce a large friction force and reduce the working range to less than 2 μm. Therefore, as suitable signals for the electrostatic actuator, the duration of the voltage to be applied to the electrodes must be short enough so as not to influence the working range.

4.2. Driving signal for full working range

As described in Section 4.1, the long period for the applied voltage to the actuator greatly reduces the working range. Thus, the impulse signal is introduced as the applied signal, which does not influence the working range and enables the actuator for full working range motion. The duration of the applied sig-

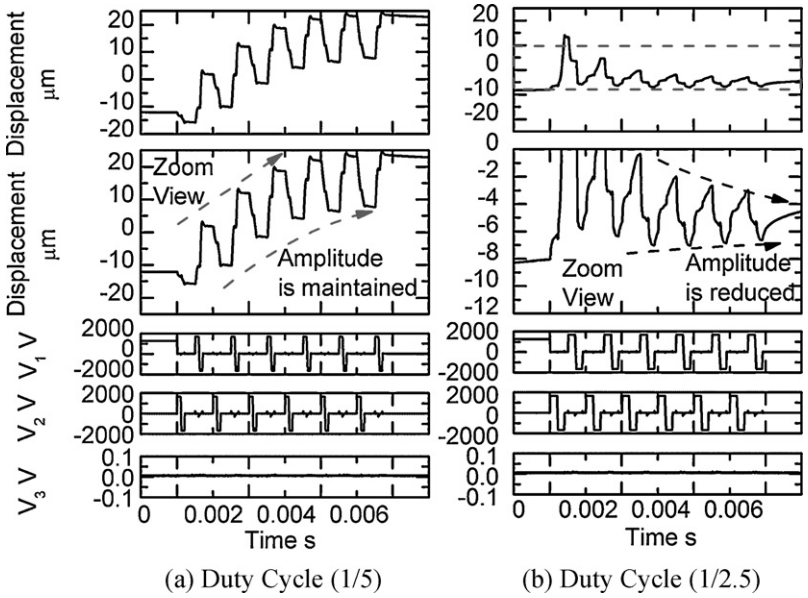


Fig. 8. Open-loop displacement characteristics with the impulse signal and the 10-mPa s silicone oil. (a) Duty cycle (1/5) and (b) duty cycle (1/2.5).

nal for the non-zero voltage is shorter than that of the normal signal for reducing the effect of the attractive force between the electrodes.

Two impulse signals, the duty cycles of which are (1/5) and (1/2.5), are used in the experiments. Both signals have a period of 1 ms. Fig. 8(a) shows the open-loop displacement characteristic with an impulse signal for which the duty cycle is (1/5) and the 10-mPa s silicone oil. This figure shows that the initial displacement is almost the same as that using the normal signal (see Fig. 7). However, the amplitude of the displacement is maintained at approximately 20 μm_{p-p}. This indicates that this impulse signal does not influence the working range of the actuator. In addition, the displacement is shifted upward. The reason for this is considered to be that the frictional effect depends on the motion direction of the actuator. Fig. 8(b) shows the open-loop displacement response with the impulse signal for which the duty cycle is (1/2.5) and the 10-mPa s silicone oil. In comparison with the duty cycle (1/5), the impulse signal for which the duty cycle is (1/2.5) decreases to approximately 2 $\mu m_{p-p.}$ Therefore, it can be concluded that the impulse signal for which the duty cycle is (1/5) is suitable for driving the mover in the full working range.

5. Control performance

In this section, the control performance of the electrostatic actuator supported by lubricating oil is examined in order to demonstrate the effectiveness of the actuator. The control system was designed based on a PID controller so that the system had high precision positioning performance. Fig. 9 shows a block diagram of the control system, including a linearizer and a driving signal unit. For this mechanism, the thrust force produced by the electrostatic actuator is proportional to V^2, which is not appropriate for high-precision motion control. Therefore, in this control system, a linearizer unit is added in order to cancel the nonlinear characteristics. The driving signal unit translates the output of the linearizer to the impulse signal. The tuned PID controller parameters are $K_p = 25$, $K_i = 7.5$, and $K_d = 15$.

In the experiments, an impulse signal with a duty cycle of (1/5) is used. Fig. 10 shows the experimental positioning results obtained using periodic rectangular inputs for a step height of 10 μm and a 1-ms control period. In the present study, each experiment is repeated 20 times for all of the positioning performances. The maximum value of the average positioning error is less than 16 nm. Figs. 11 and 12 show the experimental position-

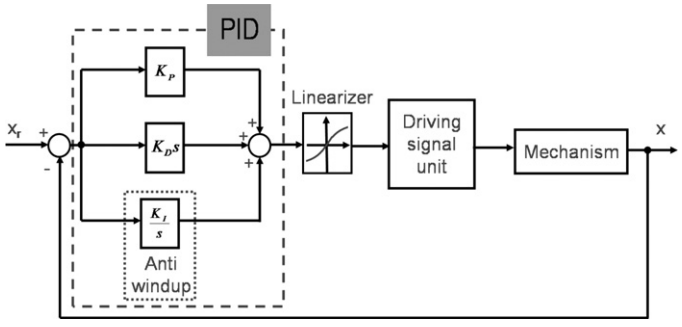


Fig. 9. Block diagram for the control system based on the PID controller.

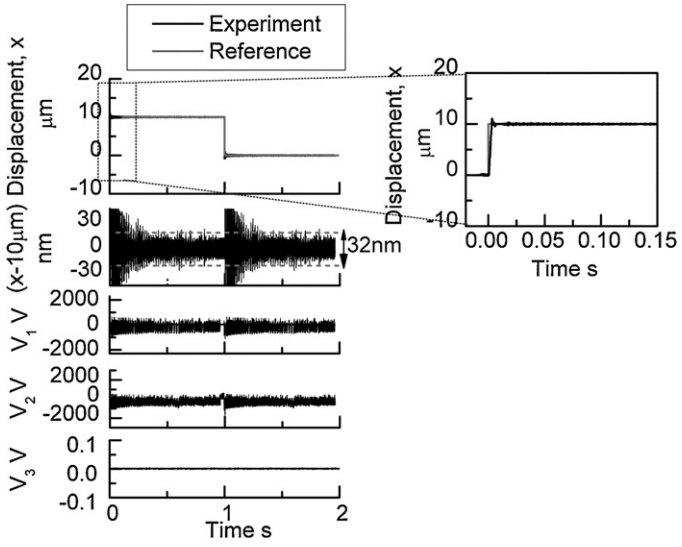


Fig. 10. Positioning results obtained using periodic rectangular inputs for a step height of 10 μm and a control period of 1 ms.

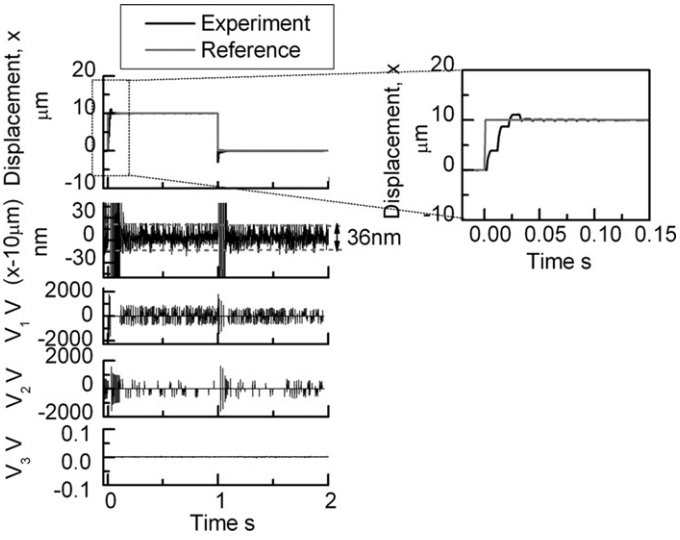


Fig. 11. Positioning results obtained using periodic rectangular inputs for a step height of 10 μm and a control period of 10 ms.

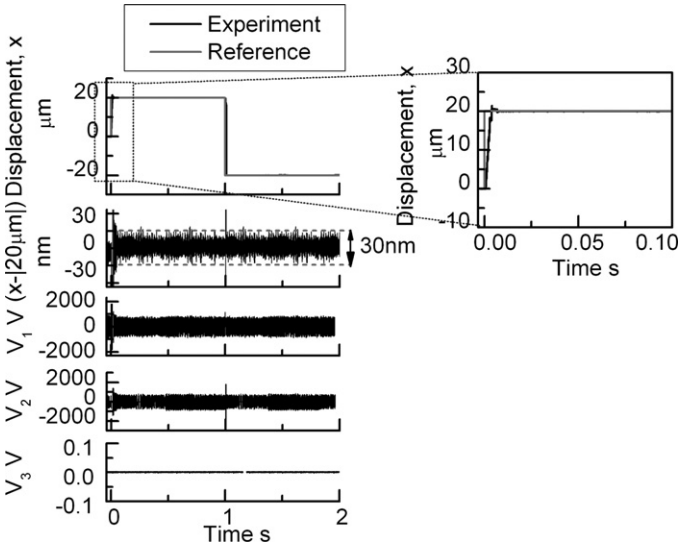


Fig. 13. Positioning results obtained using periodic rectangular inputs for a step height of ±20 μm and a control period of 0.5 ms.

ing responses to periodic rectangular inputs for a step height of 10 μm and control periods of 10 and 0.5 ms, respectively. These figures show that the maximum average positioning error is less than 18 nm, and the duty cycle of (1/5) is useful for the control system even though the control period is changed. Moreover, the shortest control period, i.e., 0.5 ms, yielded a smaller maximum average positioning error, which is less than 15 nm. In addition, the 0.5-ms control period exhibits a faster rise time. Fig. 13 shows the step response of the actuator for an input of up to 40 μm. The mover reaches the reference position quickly and smoothly. The average positioning error is less than 15 nm. These results indicate that the impulse signal enables full working range motion of the actuator.

A stepwise response was examined in order to clarify the positioning resolution performance of the actuator. Fig. 14 shows the response of the actuator to a stepwise input for a step height of 30 nm and a 0.5-ms control period. The positioning resolution is less than 15 nm under the designed control system.

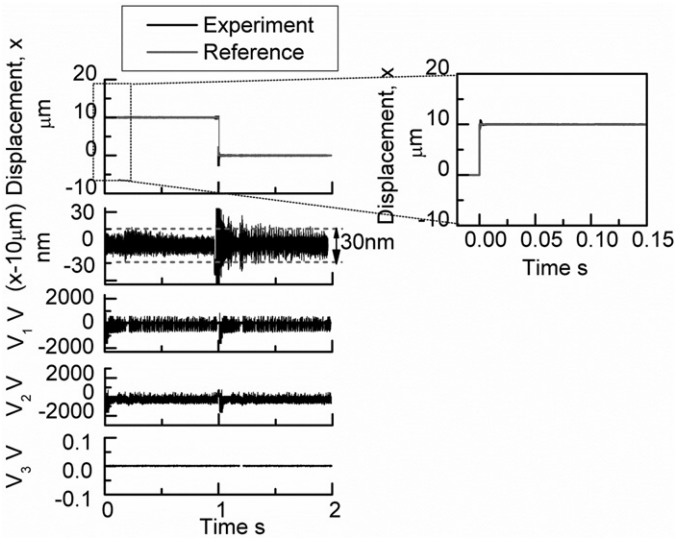


Fig. 12. Positioning results obtained using periodic rectangular inputs for a step height of 10 μm and a control period of 0.5 ms.

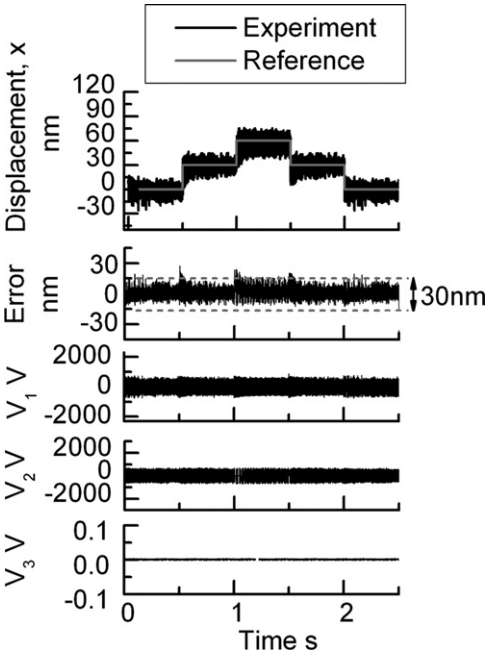


Fig. 14. Positioning resolution obtained using stepwise inputs for a step height of 30 nm and a control period of 0.5 ms.

6. Conclusions

Electrostatic actuators supported by only lubrication oil have significant advantages such as the ease of fabrication for a simple structure without hinges and precise ball guides, and the ease of maintenance without the use of precise balls. The previous electrostatic actuators do not have these advantages. However, the characteristics of the actuator were not clear. Thus, the present paper examined the basic open-loop characteristics of an electrostatic actuator supported by only lubricating oil using the electrostatic actuator with two mover layers. The oil viscosity was selected as the factor influencing the lubrication characteristics and its influences were examined. The experimental results indicate that the longer period of applying voltages greatly decreases the produced thrust force and the working range of the actuator.

The viscosity clearly influenced the mover motion. However, the advantage of the viscosity on the maintenance of the lubrication characteristics was not observed. For overcoming the problems caused by the applied voltages, the suitable driving method with the impulse signal was proposed and its effectiveness was validated experimentally. After that, the PID controller with the suitable impulse signal was designed to demonstrate the effectiveness of the actuator. The experimental results indicate that the control system has the full working range motion over 40 μm and the positioning resolution of better than 15 nm. This paper focused on the examination of the basic and qualitative characteristics in which the significant characteristics of the actuator have been clarified. The actuator structure and component materials of the experimental actuator were not optimized. However, the experimental actuator having the simple structure which shows the high precision positioning performance.

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